

VIJENDER SINGH

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SUMMARY

Bioinformatics leader with 20+ years delivering multi-omics solutions to clinicians, pharma and biotech clients - from experimental design to clinical translation - with a track record of \$12m+ in secured funding and an 80% client retention rate.

SKILLS

Multi-omics analysis: RNA-seq, ChIP-seq, ATAC-seq, single-cell (CITE-seq, ATAC-seq), genome/transcriptome assembly, variant detection

Languages & tools: Python, R, Shell scripting, Nextflow, Docker, Singularity, cloud compute (AWS S3), iRODS, high-performance computing

Leadership: Team building and mentoring, strategic planning, stakeholder communication, business development, financial planning

EXPERIENCE

Lead Bioinformatician, Software Development — UK Health Security Agency *November 2024 – Present*

- Led development of Nextflow pipelines for three new pathogens, expanding surveillance capability.
- Pipelines are typically accredited by UKAS against ISO 15189:2022, ensuring clinical safety and data reliability.
- Recruited and led a team of three to deliver pipeline support for clinical laboratories.
- Built an automated data archiving application, integrating iRODS with AWS S3.
- Established Singularity container standards, improving consistency across the bioinformatics division.
- Applied ML to infer antimicrobial resistance phenotypes from TB whole-genome sequencing data, supporting clinical surveillance decisions.

Biotech Industry & Pharma Engagement Consultant

- Delivered end-to-end single-cell and multi-omics support for Frequency Therapeutics, contributing to a hearing-restoration drug candidate (FX-322) that advanced to Phase 1/2 clinical trials.
- Nicholas Piramal India Ltd. : Consulted on identification of antifungal gene targets in *Candida albicans*, using microarray transcriptomics to track gene expression changes in oral fungal infection samples from head and neck radiotherapy patients.
- Provided transcriptomics and multi-omics pipeline support for RNAi therapeutics developer Drishti Discoveries, from experimental design through target discovery and biomarker identification.

Associate Director, Computational Biology Core & Clinical Bioinformatics Consultant — UConn Health *October 2020 – October 2024*

- Built and led a bioinformatics team supporting three regional University of Connecticut campuses, delivering 50+ genomics projects with an 80% repeat-collaboration rate.
- Secured approximately \$12 million in grant funding at a 45% proposal success rate, guiding experimental design and analytical methodology.
- Generated approximately \$70,000 in annual revenue (FY2022–2024) through client consultation on project feasibility and data analysis.
- Presented complex findings to stakeholders and researchers of varied technical backgrounds; promoted core services at conferences and trade fairs.

Lead Bioinformatics Scientist, Computational Biology Core — University of Connecticut *October 2016 – October 2020*

- Designed scalable analytical workflows for transcriptomics, ChIP-seq, and single-cell genomics.
- Established core-to-client communication and project tracking processes across research teams.

Senior Research Associate / Postdoctoral Fellow — University of Dundee *May 2007 – October 2016*

- Directed independent chromatin remodeling research projects, coordinating national and international collaborations.
- Published 10 peer-reviewed articles in international journals; supervised PhD students and trainees.

EDUCATION

PhD, Transcriptional Regulation and Galactose Metabolism — Indian Institute of Science (1999–2007)

Master's Degree, Biochemistry — Bangalore University

BS, Chemistry, Botany and Zoology — University of Rajasthan

CERTIFICATIONS

AgilePM® v3 Practitioner | AgilePM® v3 Foundation

Data Science with Machine Learning & AI Career Accelerator — University of Cambridge (in progress)